

FitHub AI

“Train smarter. Live better.”

Comprehensive Software Project Life Cycle Document

Project Name	FitHub AI
Application Type	AI-Powered Fitness Coaching Web Application
Primary Tech Stack	Python (FastAPI), Jinja2, SQLAlchemy, SQLite, Google Gemini API, Docker, Uvicorn

6. Project Testing Phase

Verification & Testing Methods

- Local HTTP Verification:** Executed app locally at `http://127.0.0.1:8012` and confirmed an HTTP 200 OK status.
- Functional Form Integration:** Evaluated multi-field submissions in the Workout Planner under various user conditions (e.g., 10-minute session, no equipment, fat loss goal).
- AI Output Quality Check:** Verified that AI outputs accurately generated structured JSON/card blocks containing calories burned, exercise categories, and durations.
- Deployment Testing:** Executed dry runs of `deploy_vps.sh` to confirm virtual environment activation (`.venv`), dependency installation via `pip`, and dynamic `.env` generation.